Inertia and balanced forces

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Predict motion when the net force is zero.

SUCCESS CRITERIA - I can...

1. Add forces to decide whether they balance.
2. Predict constant velocity when net force is zero.
3. Explain why changing direction requires net force.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “net force” and “inertia” to explain your answer. Sentence starters are optional.

net force: the vector sum of all forces on one object

inertia: the tendency to keep the same velocity without net force

balanced: adding to zero net force in this force context

01 Warm-up: bring back an earlier idea

A=(**6**, **-4**) m and B=(**-4**, **8**) m.
1. **Find R**, magnitude and direction.
2. **Choose C** so A+B+C=**0** and **sketch the route**.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: ΣF = **0** ⇒ a = **0** ⇒ v = constant.
ΣF means net force. Balanced forces allow rest or steady motion in a straight line.

Worked reasoning: A book has **6 N** upward support and **6 N** downward weight. Net force is zero. Its velocity stays constant; it could be resting or moving steadily.

Your turn: Draw the two force arrows. Then **sketch two possible velocity-time graphs** consistent with this balance.

Sentence starter: In my work, I used … to show … . I labeled it … .



Book: compare support and weight.

Marzano check - circle: ① ② ③

03 Infer the motion constraint

A **4 kg** cart experiences **6 N** right and **6 N** left. It currently moves at **-3 m/s**. **Find acceleration** and its displacement during the next **5 s**.

Sentence starter: I found … [unit]. This value means … in this situation.



The dot represents the object. Velocity: 3 m/s left

Marzano check - circle: ① ② ③

04 Inertia in a braking bus

1. **Explain a passenger’s forward motion** relative to a braking bus using an approximately inertial ground frame.
2. **Name the external interaction** that changes passenger velocity.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: ① ② ③

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Continue down each column.

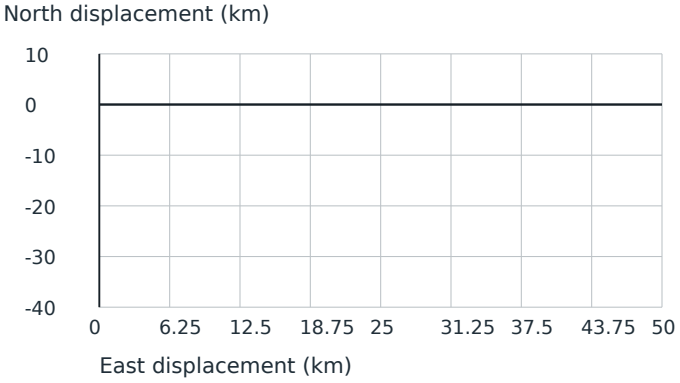
Sentence starters are optional. You may plan in any language.

05 Engine-off spacecraft

A spacecraft coasts in an ideal region with zero net force. Its velocity is (**4**, **-3**) km/s, with *east* +x and *north* +y. **Draw the displacement over 10 s**, find its magnitude, and **explain why an engine** is not needed to maintain this motion.

Word help: ideal = using stated simplifying conditions

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .



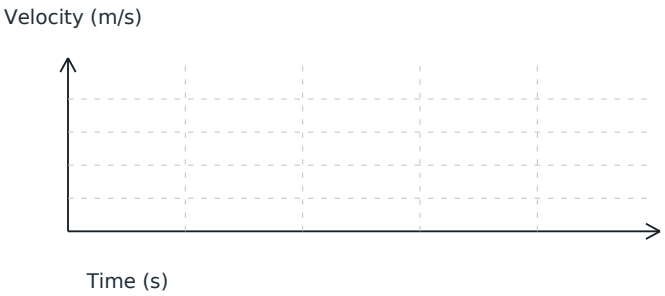
Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A car starts at **+15 m/s** and brakes at **-3 m/s²** until rest. A student extends that acceleration to **7 s**. **Find the stop time**/distance and **explain why extension** is invalid for ordinary braking.
Sketch the velocity–time graph; **label the given values**.

Word help: acceleration = rate of change of velocity

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

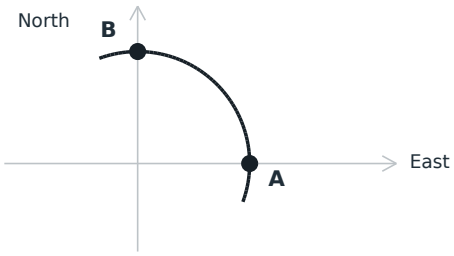


Marzano check - circle: ① ② ③

07 Exit check

A runner goes from A toward B around the shown curve at constant speed. A classmate says this is equilibrium because speed does not change. **Draw velocity arrows** at both points and use them to correct the statement.

Sentence starter: In my work, I used … to show … . I labeled it … .



Path sketch: runner moves from A toward B.

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

An elevator moves upward at a steady **2 m/s**. Its mass is **500 kg**; use g = **10 N/kg**. **Complete its force diagram** and **find cable tension**. Then **sketch a case** with lower tension while it still moves upward. **Describe how its speed changes**.

Sentence starter: I found … [unit]. This value means … in this situation.

Add and label force arrows.

Choose the elevator as the object.

Marzano check - circle: ① ② ③